

Efficacy and safety of novel airway balloon cryoablation system for malignant central airway obstruction: a prospective, multicentre, randomised, non-inferiority study

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ABSTRACT

Background A novel airway balloon cryoablation (ABC) system using liquid nitrogen as cryogen was developed for bronchoscopic intervention in malignant central airway obstruction (MCAO). This study aimed to evaluate the efficacy and safety of this system.

Methods This was a prospective, randomised, controlled, non-inferiority study that enrolled MCAO patients at 15 sites in China. Patients were assigned to the ABC group and the ERBOKRYOCA Cryosurgical (EC) system group in a 1:1 ratio. Airway tumour debulking with multimodality was permitted before cryoablation. The primary outcome was the airway patency rate after 6 weeks of intervention with a non-inferiority margin of –10%. Secondary outcomes included modified Medical Research Council (mMRC) Dyspnoea Scale and Karnofsky Performance Scale (KPS) assessment, and the duration of cryoablation.

Results 198 patients were randomised. After 6 weeks of intervention, the airway patency rate was 78.49% in the ABC group and 60.92% in the EC group, showing that the difference was over the non-inferiority margin of –10% at 17.58% (95% CI 4.35% to 30.80%), $p < 0.001$. The mMRC and KPS scores were significantly improved after 3 and 6 weeks in both groups, with no difference. The duration of cryoablation was shortened remarkably in the ABC group (378.29 ± 399.54 s vs 624.93 ± 443.72 s, $p < 0.001$). The prevalence of bleeding was similar in the two groups, without life-threatening events.

Conclusions The ABC system provided non-inferior and subsequent superior effect in airway patency rate compared with traditional carbon dioxide-driven cryoprobe in MCAO. This study supports this novel system as an alternative to cryoablation via bronchoscope for MCAO patients.

Trial registration number ChiCTR2100042051.

INTRODUCTION

Patients with malignant central airway obstruction (MCAO) usually associate with poor performance status and suffer from life-threatening respiratory distress that requires immediate assessment and intervention. With the rapid development of

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Traditional flexible carbon dioxide (CO₂) or N₂O-driven cryoprobe is the commonly used device in the airway cryoablation via bronchoscope in malignant central airway obstruction (MCAO) for decades.

WHAT THIS STUDY ADDS

⇒ As a part of multimodal interventions in MCAO management, the novel airway balloon cryoablation (ABC) system demonstrated non-inferior effect in airway patency rate compared with the traditional cryoprobe. In addition, the operation time was significantly reduced, and safety was comparable.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This novel ABC system provides a promising alternative to traditional CO₂-driven cryoprobe in bronchoscopic cryoablation for MCAO patients.

interventional pulmonology, therapeutic bronchoscopy has been confirmed as an effective method for the treatment of MCAO cases, enabling the recanalisation of airway stenosis, relieving symptoms, prolonging survival and providing opportunities for subsequent anti-tumour therapy.^{1 2} Mechanical debulking, airway dilation techniques, thermal ablation techniques and cryoablation are commonly used modalities. MCAO is classified into endoluminal (intrinsic), extraluminal (extrinsic) and mixed type (combined intrinsic and extrinsic). Different therapeutic approaches should be chosen based on the individual characteristics of airway stenosis. Frequently, a multimodality strategy is required to achieve airway restoration.^{3 4}

Cryoablation is a unique thermal therapy that uses extremely low temperatures and freeze-thaw cycles to induce multiple forms of cell injury. The direct mechanism includes formation of ice crystals, resulting in organelle damage and transcellular fluid shifts. Ischaemia caused by local vasoconstriction



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and thrombosis, and potential immune-mediated factors leading to delayed cell death, will last for weeks.⁵ Compared with other thermal therapies, cryoablation is friendly to cartilage, airway stents and pacemakers, and not associated with electrical accidents.⁶ Since its first use in endoluminal airway tumours in 1968, cryoablation has demonstrated reliable efficacy and safety in treating MCAO via bronchoscope.⁷ Previous studies have shown that a faster cooling rate, a lower target temperature, a longer freezing time and a slower thawing rate enhance the efficacy of cellular injury.⁸

Cryoprobes, based on the Joule-Thomson physical principle, available in diameters of 1.1–2.4 mm, can pass through the working channel of a bronchoscope. When performing cryoablation for an airway lesion, two to three freeze-thaw cycles, with 20–30 s of freezing time per cycle, are required for a single point. To cover the entire surface of the lesion, the operation time is inevitably prolonged.⁹ Here, we introduce a novel cryoablation system with an airway balloon catheter, driven by a stable liquid nitrogen delivery system. This system increases the freezing working area and achieves a lower target temperature, enhancing the cryoablation effect and potentially reducing operation time. The study aims to compare the efficacy and safety of this novel airway balloon cryoablation (ABC) system with the traditional carbon dioxide (CO₂)-driven cryoprobe in the treatment of MCAO.

STUDY DESIGN AND METHODS

Participants and study design

This study was designed as a multicentre, prospective, randomised, controlled, non-inferiority trial. Patients enrolled from 15 hospitals in China between April 2021 and December 2023. Patients were randomly assigned to the ABC group and the ERBOKRYOCA Cryosurgical (EC) system group in a 1:1 ratio. Major inclusion criteria were: (1) patients aged between 18 and 75 years old; (2) central airway obstruction due to malignant tumours; (3) airway stenosis affecting more than 50% of the airway lumen and (4) intrinsic or mixed type airway obstruction. Patients meeting any of the following criteria were excluded: (1) extrinsic compression stenosis requiring stent placement; (2) multiple sites of airway involvement; (3) severe obstruction with an airway lumen diameter of less than 4 mm; (4) radiation therapy at the site of airway obstruction within 3 months prior to enrolment; (5) simultaneous targeted therapy with EGFR, ALK or ROS1 inhibitors without disease progression; (6) contraindications for bronchoscopy or anaesthesia; (7) a life expectancy of

less than 3 months and (8) pregnancy, lactation or other conditions deemed unsuitable by the researchers based on the patients' potential benefit. The study has been registered on the Chinese Clinical Trial Registry (ChiCTR2100042051) and the protocol is accessed to: <https://www.chictr.org.cn/showproj.html?proj=120130>.

Cryoablation systems

The ABC system comprises a cryoablation console and an airway cryoablation catheter (SENSCURE Biotechnology Co., Ningbo, China, [figure 1](#)). The catheter, in conjunction with the console, is used for the cryoablation of airway lesions. Liquid nitrogen serves as the cryogen, circulates within a closed-loop system, with no external leakage or requirement for passive ventilation. The catheter can be introduced into the airway through the working channel of the bronchoscope, with the distal end equipped with an inflatable balloon of various sizes (diameter ranging from 4 to 12 mm, length ranging from 5 to 30 mm). When the catheter reaches the airway lesion, the balloon will be inflated to ensure close contact with the lesion. Then, liquid nitrogen can be delivered rapidly from the console via the catheter into the balloon with an internally circulated manner, thereby achieving cryoablation. A temperature of $-145^{\circ}\text{C} \pm 10^{\circ}\text{C}$ can be achieved at the surface of the balloon. Cryoablation in the EC group was performed with the EC system (Erbe Elektromedizin, Tübingen, Germany) through a 1.9 mm or 2.4 mm diameter cryoprobe using CO₂ as the cryogen. The temperature of the probe tip is around -35°C to -50°C .

Bronchoscopic procedures

All the bronchoscopic therapeutic procedures were performed under general anaesthesia with flexible and/or rigid bronchoscopy by practiced bronchoscopists at each site. Before cryoablation, patients were assessed bronchoscopically and underwent airway tumour debulking with multimodality to achieve initial airway patency according to consensus and guidelines on MCAO management.^{4 10 11} Modalities can include mechanical debridement with rigid bronchoscopy, thermal therapy, cryo-canalisation, snare and forceps and were chosen depending on clinical settings. After debulking, a different device was inserted through the working channel of a therapeutic flexible bronchoscope according to the randomisation to perform cryoablation. For the ABC group, the size of the balloon was chosen based on the airway lumen diameter. The cryoablation involves two

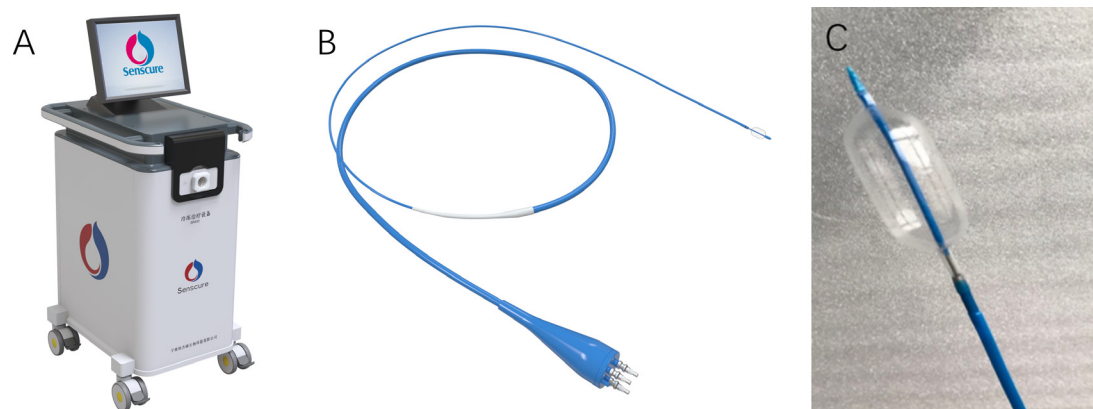


Figure 1 The novel airway balloon cryoablation system. (A) The console of the system; (B) The catheter which can be connected to the cryoablation system and deliver circulating liquid nitrogen to the distal balloon. (C) The inflated balloon.

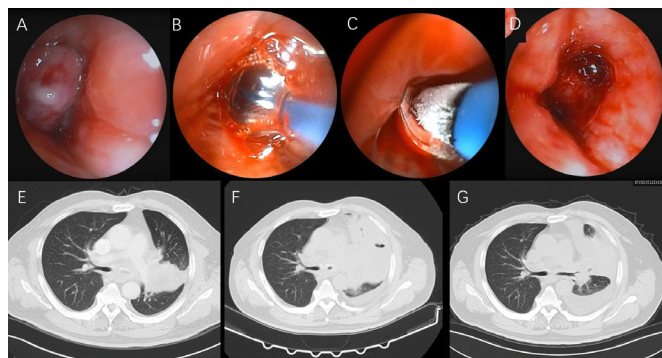


Figure 2 A patient treated with airway balloon cryoablation system and CT scan follow-up. The patient presented with worsening dyspnoea and a mass at the left hilum was revealed on CT scan. A therapeutic bronchoscopy was performed for interventional treatment and pathological diagnosis which was confirmed as lung adenocarcinoma (stage IVB with ROS-1 fusion). (A) The left main bronchus was obstructed with neoplasm; (B) The balloon was inserted and inflated after tumour debulking; (C) Cryoablation with the balloon filled with liquid nitrogen; (D) The left main bronchus after the intervention; (E) CT scan before the intervention; (F) CT scan at 3 weeks after the intervention; (G) CT scan at 6 weeks after the intervention.

cycles of freezing and rewarming (figure 2). For the EC group, the cryoprobe was used at multiple points to ensure the coverage of the airway lesion and one to three freeze-thaw cycles were performed for each point. Each cycle consisted of 45–60 s of freezing and complete thawing (figure 3). Before discharge, if necessary, bronchoscopy should be performed again because the devitalised tissue in the airway might develop oedema and necrosis later, which would cause further narrowing of the airway.

Outcomes and data collection

The primary outcome was the airway patency rate after 6 weeks of intervention. The secondary outcomes included modified Medical Research Council (mMRC) scale¹² and Karnofsky

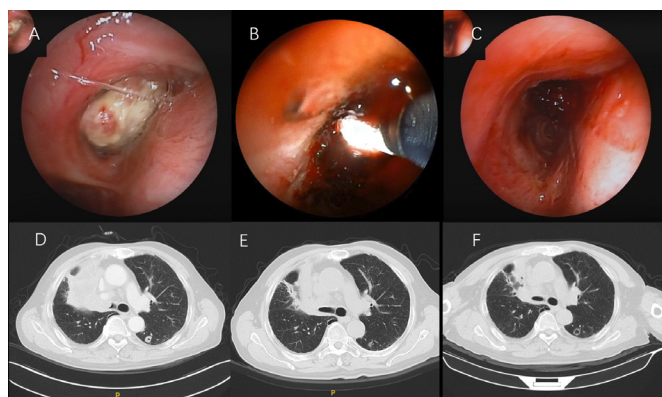


Figure 3 A patient treated with ERBOKRYOCA Cryosurgical system and CT scan follow-up. A patient diagnosed with squamous cell carcinoma, stage IIIB, suffered from disease progression after multiple regimens including chemotherapy, radiotherapy and immunotherapy. (A) The right main bronchus was completely obstructed; (B) Cryoablation with traditional cryoprobe after tumour debulking; (C) The right main bronchus after the intervention; (D) CT scan before the intervention; (E) CT scan at 3 weeks after the intervention; (F) CT scan at 6 weeks after the intervention.

Performance Score (KPS)¹³ at 3 and 6 weeks after the intervention, and the duration of cryoablation. The immediate success rate of cryoablation and device performance was also recorded and evaluated by investigators. Any adverse events related to the cryoablation system or interventional procedure were reported for the safety assessment.

Demographics, medical history and vital signs of each patient were collected, and laboratory tests and chest CT scan were also performed before the bronchoscopy to further confirm the patients' suitability for the study. Postintervention visits were scheduled for 3 and 6 weeks. Assessments of symptoms and functions, and adverse events were gathered via interview, and CT scans were followed as well. The airway lumen diameter was measured by independent radiologists. The degree of airway obstruction was defined as the percentage of the reduced lumen diameter caused by tumour to the normal. Airway patency was defined as the restoration of airway lumen diameter to >50% of normal anatomical diameter after intervention. The airway patency rate was calculated as the proportion of patients demonstrating sustained airway patency (≥ 6 weeks) following the intervention.

Statistical analysis

The sample size was calculated based on the primary outcome. According to the existing clinical evidence^{14 15} and the experience of clinical experts, the expected airway patency rate after therapeutic bronchoscopy of both groups could be 95% and the non-inferiority margin was set at -10% . Using a one-sided α of 0.025, a power of $1-\beta$ at 80% and a drop out of 20%, about 200 patients were planned to be enrolled in this study. All statistical analyses were performed using SAS V.9.4 (SAS). Categorical data were presented as number (%), and continuous data were presented as mean (SD) or median (IQR) according to whether they were normally distributed. Categorical variables were compared with the Fisher's exact test or χ^2 test. The ordinal data were analysed using the Wilcoxon rank-sum test or Cochran-Mantel-Haenszel test. Comparisons of quantitative data were conducted by independent t-test or Wilcoxon rank-sum test according to the homogeneity of variance and normal distribution. For the primary outcome, statistical analysis will be conducted at a one-sided significance level of 0.025. Two-tailed p values of less than 0.05 were considered statistically significant differences for other indicators.

RESULTS

Patient and disease characteristics

Between April 2021 and December 2023, 233 patients with MCAO were screened, and 198 patients who met the inclusion/exclusion criteria were included and randomised ultimately. Among them, 100 patients were enrolled in the ABC Group and 98 patients in the EC group (figure 4). Table 1 shows the baseline characteristics of all the patients and their disease information. There was no difference in age, gender, body weight or height between the two groups. Primary lung cancer was identified as the most common aetiology of MCAO. Other MCAO cases were caused by metastasis or invasion of tumours derived from oesophagus, thyroid, mediastinum, liver, kidney and other sources. The histology type and TNM staging were similar in each group. Some cases suffered from MCAO due to the progression of primary carcinoma, and the percentage of patients with previous history of antitumour therapy such as surgery, medication and radiotherapy was shown with no significant difference

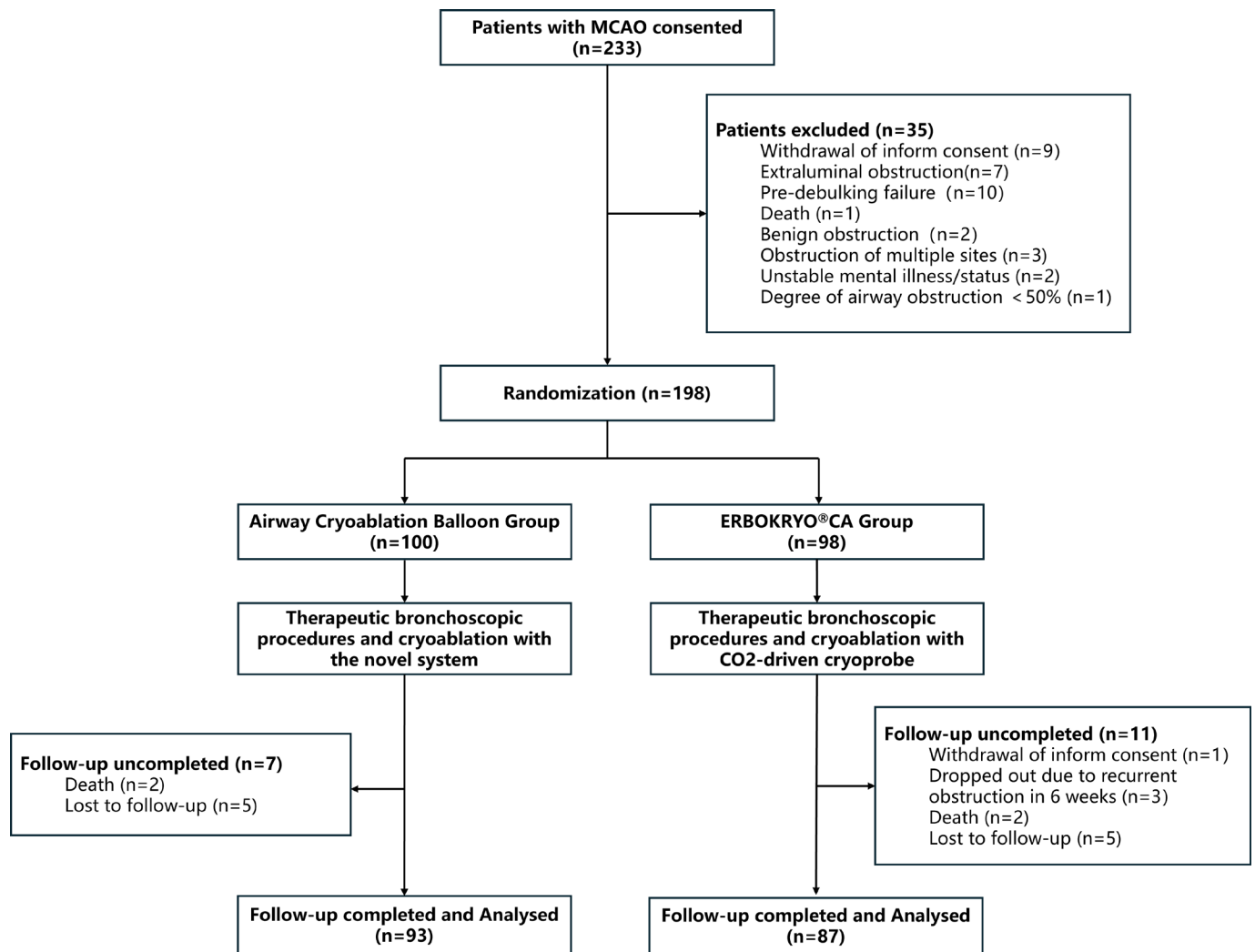


Figure 4 Patient flow. CO2, carbon dioxide; MCAO, malignant central airway obstruction.

between the two groups. About 20% of patients had received therapeutic bronchoscopy therapy before.

Each patient was evaluated through physical examination, blood tests, CT scans and bronchoscopy to rule out contraindications before randomisation and further procedures. No difference was observed between the two groups in the vital signs, site and type of airway obstruction, except that the diastolic pressure was slightly higher in the EC group (table 1). The most common site of MCAO was the right main bronchus (30.3%), followed by the left main bronchus (26.27%) and trachea (17.18%). The type of obstruction in most cases was intrinsic. The degree of airway obstruction measured on CT scan before the intervention was similar in the ABC and EC groups ($80.30\% \pm 18.79\%$ vs $79.59\% \pm 17.40\%$), as well as the diameter of the reduced and normal airway of each group (table 2).

After the interventional procedures, 7 patients in the ABC group and 11 patients in the EC group did not complete the follow-up or withdrew from the study. Among them, both groups had two patients who died due to disease progression (not related to MCAO), and five patients were lost to follow-up. In the EC group, one patient withdrew consent, and three patients experienced restenosis at the original site of the airway within 6 weeks, which indicated that they met the primary outcome in advance and treatment failure. Consequently, 93 patients in the ABC group and 87 patients in the EC group completed

follow-up and were included in the outcome analysis (figure 4). The characteristics of the patients who did not complete the study were not significantly different from those who completed (online supplemental table 2).

Cryoablation procedure characteristics and safety

The airway obstructions were recanalised, and subsequent cryoablation was performed in all the patients successfully enrolled. Most patients achieved airway restoration, with the degree of airway obstruction being less than 25% at the end of the intervention (98.00% vs 94.90%). The duration of cryoablation in the ABC group was markedly shorter than that in the EC group (378.29 ± 399.54 s vs 624.93 ± 443.72 s, $p < 0.001$).

No immediate severe adverse effects were reported. Mild bleeding was observed in 25 (25.00%) and 20 (20.41%) patients in each group, respectively, and 1 (1.02%) patient in the EC group suffered from moderate bleeding requiring medication instillation. Haemoptysis after intervention occurred in 4 (4.00%) patients of the ABC group and 1 (1.02%) of the EC group, and all cases were mild, manifesting as blood-tinged sputum (< 5 mL/24 hours) that spontaneously resolved in 48 hours without any interventions. Delayed airway restenosis due to tumour necrosis caused by cryoablation occurred in two patients in the ABC group

Table 1 Patient and disease characteristics

	ABC group (n=100)	EC group (n=98)	P value
Age (years), median (IQR)	63.50 (58.00, 70.00)	65.00 (58.00, 71.00)	0.308
Sex, female/male	81/19	83/15	0.491
Height (cm), median (IQR)	169.00 (163.00, 172.00)	169.00 (162.00, 172.00)	0.851
Weight (kg), mean (SD)	61.11±10.98	61.85±11.73	0.648
Previous radiotherapy, n (%)			
Yes	18 (18.00)	13 (13.27)	0.359
No	82 (82.00)	85 (86.73)	
Previous antitumour medication, n (%)			
Yes	83 (83.00)	77 (78.57)	0.429
No	17 (17.00)	21 (21.43)	
Previous surgery, n (%)			
Yes	51 (51.00)	58 (59.18)	0.247
No	49 (49.00)	40 (40.82)	
Previous interventional bronchoscopic therapy, n (%)			
Yes	19 (19.00)	20 (20.41)	0.803
No	81 (91.00)	78 (79.59)	
Vital signs before intervention			
Respiratory rate (times per minute), median (IQR)	19.00 (18.00, 20.00)	19.00 (18.00, 20.00)	0.815
Heart rate (beats per minute), median (IQR)	80.00 (75.00, 86.00)	80.00 (76.00, 87.00)	0.498
Temperature (°C), median (IQR)	36.40 (36.30, 36.60)	36.50 (36.30, 36.60)	0.114
Systolic pressure (mm Hg), median (IQR)	120.00 (112.00, 131.00)	122.00 (114.00, 129.00)	0.794
Diastolic pressure (mm Hg), median (IQR)	73.00 (70.00, 80.00)	77.60 (71.00, 81.00)	0.031
Tumour histology, n (%)			0.718
Primary lung cancer	79 (79.00)	80 (81.63)	
Oesophageal carcinoma	1 (1.00)	0 (0.00)	
Thyroid carcinoma	2 (2.00)	1 (1.02)	
Mediastinal tumour	2 (2.00)	0 (0.00)	
Others*	16 (16.00)	17 (17.35)	
Lung cancer pathology, n (%)			0.877
Small cell lung cancer	9 (9.00)	6 (6.12)	
Non-small cell lung cancer	4 (4.00)	5 (5.10)	
Adenocarcinoma	18 (18.00)	21 (21.43)	
Squamous cell carcinoma	45 (45.00)	47 (47.96)	
Mixed histological type	1 (1.00)	0 (0.00)	
Others†	2 (2.00)	1 (1.02)	
TNM staging, n (%)			0.112
I	3 (3.00)	2 (2.04)	
II	0 (0.00)	5 (5.10)	
III	36 (36.00)	30 (30.61)	
IV	40 (40.00)	45 (45.91)	
Non-staging	21 (21.00)	16 (16.33)	

Continued

Table 1 Continued

	ABC group (n=100)	EC group (n=98)	P value
No significant differences in baseline characteristics were noted between the two groups.			
*Others tumour included colon cancer, hepatocellular carcinoma, osteoclastoma, adenoid cystic carcinoma derived from parotid gland and external auditory canal, kidney carcinoma and inflammatory myofibroblastic tumour.			
†Other pathology of lung cancer included adenoid cystic carcinoma, fibrosarcoma and neuroendocrine carcinoma.			
ABC, airway balloon cryoablation system; EC, ERBOKRYOCA Cryosurgical system.			

on the fourth and ninth day after the intervention, and they were relieved after clearing the necrotic tissue in a timely manner. The details of the bronchoscopic procedures are shown in [table 3](#).

Primary outcome

At 3 and 6 weeks postinterventional follow-up, the loss of airway diameter due to malignant tumour was remarkably improved through airway tumour debulking and cryoablation, and the degree of airway obstruction was effectively ameliorated in both groups as well ([table 4](#)). The airway patency rates after 6 weeks of intervention were 78.49% (73/93) in the ABC group and 60.92% (53/87) in the EC group, yielding an absolute between-group difference of

Table 2 Characteristics of airway obstruction

	Total (n=198)	ABC group (n=100)	EC group (n=98)	P value
Location of airway obstruction on CT, n (%)				0.718
Upper 1/3 part of trachea	18 (9.09)	9 (9.00)	9 (9.18)	
Middle 1/3 part of trachea	5 (2.53)	3 (3.00)	2 (2.04)	
Lower 1/3 part of trachea	11 (5.56)	6 (6.00)	5 (5.10)	
Carina	1 (0.51)	1 (1.00)	0 (0.00)	
Right main bronchus	60 (30.3)	26 (26.00)	34 (34.69)	
Right bronchus intermedius	24 (12.12)	11 (11.00)	13 (13.27)	
Left main bronchus proximal	23 (11.62)	19 (19.00)	4 (4.08)	
Left main bronchus distal	29 (14.65)	10 (10.00)	19 (19.39)	
Lobar bronchus	42 (21.22)	24 (24.00)	18 (18.36)	
Obstruction type, n (%)				0.927
Intrinsic	140 (65.66)	71 (71.00)	69 (70.40)	
Mixed	58 (34.34)	29 (29.00)	29 (29.60)	
Degree of airway obstruction on CT (%), mean (SD)		80.30±18.79	79.59±17.40	0.642
Loss of airway diameter (mm), mean (SD)		9.96±3.45	10.40±3.45	0.397
Diameter of normal airway (mm), mean (SD)		12.59±3.82	13.49±4.05	0.121
The table shows no difference in the characteristics of airway obstruction between the two groups.				
ABC, airway balloon cryoablation system; EC, ERBOKRYOCA Cryosurgical system.				

Table 3 Bronchoscopic intervention procedures

	ABC group (n=100)	EC group (n=98)	P value
Anaesthesia, n (%)			0.366
General anaesthesia	99 (99.00)	95 (96.94)	
Sedation	1 (1.00)	3 (3.06)	
Tumour debulking, n (%)			0.644
Mechanical debulking	4 (4.00)	4 (4.04)	
Argon plasma coagulation	26 (26.00)	33 (33.67)	
Laser therapy	54 (54.00)	54 (55.10)	
Electrocautery	31 (31.00)	41 (41.84)	
Snare	13 (13.00)	13 (13.27)	
Other modalities*	35 (35.00)	26 (26.53)	
Duration of cryoablation (s), mean (SD)	378.29±399.54	624.93±443.72	<0.001
Bleeding during cryoablation, n (%)			0.499
Mild	25 (25.00)	20 (20.41)	
Moderate	0 (0.00)	1 (1.02)	
Haemoptysis after intervention, n (%)	4 (4.00)	1 (1.02)	0.369
Severe adverse events related to device, n (%)	2 (2.00)†	0 (0.00)	0.498
Degree of airway obstruction after intervention, n (%)			0.277
Grade I (≤25%)	98 (98.00)	93 (94.90)	
Grade II (26%–50%)	2 (2.00)	5 (5.10)	
Grade III (51%–75%)	0 (0.00)	0 (0.00)	
Grade IV (76%–90%)	0 (0.00)	0 (0.00)	
Grade V (91%–100%)	0 (0.00)	0 (0.00)	

Bronchoscopist could use multiple modalities for tumour debulking in a patient based on individual characteristics.

The table demonstrates the details on bronchoscopic intervention procedures. Most patients achieved airway restoration after intervention. The duration of cryoablation was markedly reduced in the ABC group.

*Other modalities included cryocanalisation and forceps.

†Airway obstruction deteriorated due to necrosis caused by cryoablation.

ABC, airway balloon cryoablation system; EC, ERBOKRYOCA Cryosurgical system.

17.58%. The 95% CI for this difference was 4.35%–30.80% with $p < 0.001$, indicating that the primary outcome of this study had been met (table 4). The intention-to-treat (ITT) analysis yielded consistent results (online supplemental table 3).

Secondary outcomes

After the therapeutic intervention, respiratory symptoms and function assessed by mMRC scale improved significantly in both groups before discharge, at 3 weeks and at 6 weeks ($p < 0.001$). Compared with baseline performance, the KPS scores also increased during the follow-up in both groups ($p < 0.001$). No statistical difference was observed between the two groups in the comparison of symptomatic and functional assessments (table 5). ITT analysis also achieved similar results (online supplemental table 4).

Performance evaluation of ABC system

The performance evaluation of the novel ABC system was subjectively appraised by bronchoscopists who performed the cryoablation. The items included convenience, stability, rationality, reliability and safety of the console, catheter and balloon. The scale was classified into five grades: poor, relatively poor, moderate, good and excellent. More than 99% of bronchoscopists regarded the ABC system as good or excellent (online supplemental tables 5 and 6).

DISCUSSION

In this study, we demonstrated that cryoablation in MCAO using a novel ABC system with nitrogen as the cryogen is both effective and safe. Following multimodal tumour debulking, subsequent ABC system for cryoablation of residual lesions demonstrated favourable efficacy in maintaining airway patency. Compared with the conventional CO₂ cryoprobe, this novel system is non-inferior or even superior in maintaining the airway patency, and it also shortened the duration of cryoablation. Respiratory symptoms and functional performance were improved during the 6-week follow-up. The most frequent adverse events were

Table 4 Follow-up and comparison of airway obstruction parameters on CT scan

	ABC group (n=93)	EC group (n=87)	P value or 95% CI
Airway patency after 6 weeks of intervention, n (%)			(4.35% to 30.80%), <0.001
Yes	73 (78.49)	53 (60.92)	
No	20 (21.51)	34 (39.08)	
Degree of airway obstruction on CT (%), mean (SD)			
Baseline	79.15±18.76	80.61±17.00	0.718
3 weeks after intervention*	25.68±24.92	28.23±32.70	0.999
6 weeks after intervention*	19.20±22.96	22.54±29.09	0.852
Loss of airway diameter (mm), mean (SD)			
Baseline	9.79±3.45	10.38±3.52	0.276
3 weeks after intervention†	3.20±3.36	3.35±3.96	0.841
6 weeks after intervention†	2.24±2.83	2.96±4.17	0.725
Diameter of normal airway (mm), mean (SD)			
Baseline	12.55±3.92	13.37±4.13	0.188
3 weeks after intervention	12.30±4.36	11.93±4.02	0.588
6 weeks after intervention	12.27±4.01	12.03±4.23	0.708

The airway patency rates after 6 weeks of intervention in the ABC group demonstrated non-inferiority and subsequent superiority compared with the EC group, indicating that the primary outcome of this study had been met.

*The degree of airway obstruction on CT was significantly improved compared with baseline, $p < 0.001$.

†The loss of airway diameter was significantly reduced compared with baseline, $p < 0.001$.

ABC, airway balloon cryoablation system; EC, ERBOKRYOCA Cryosurgical system.

Table 5 Follow-up and comparison of symptomatic and functional assessment

	ABC group (n=93)	EC group (n=87)	P value
mMRC score, median (IQR)			
Baseline	2.00 (1.00, 3.00)	2.00 (1.00, 3.00)	0.898
Before discharge*	1.00 (0.00, 2.00)	1.00 (1.00, 2.00)	0.775
3 weeks after intervention*	1.00 (0.00, 1.00)	1.00 (0.00, 1.00)	0.578
6 weeks after intervention*	1.00 (0.00, 1.00)	1.00 (0.00, 2.00)	0.273
KPS score, median (IQR)			
Baseline	80.00 (70.00, 90.00)	80.00 (70.00, 90.00)	0.542
Before discharge†	90.00 (80.00, 90.00)	90.00 (80.00, 90.00)	0.779
3 weeks after intervention†	90.00 (80.00, 90.00)	90.00 (80.00, 90.00)	0.862
6 weeks after intervention†	90.00 (80.00, 90.00)	90.00 (80.00, 90.00)	0.602

The mMRC and KPS score of the two groups were improved significantly in both groups before discharge, at 3 weeks and at 6 weeks, with no statistical difference between the two groups.

*The mMRC score was significantly improved compared with baseline, $p < 0.001$.

†The KPS score increased significantly compared with the baseline performance, $p < 0.001$.

ABC, airway balloon cryoablation system; EC, ERBOKRYOCA Cryosurgical system; KPS, Karnofsky Performance Scale; mMRC, modified Medical Research Council Dyspnoea Scale.

bleeding during the procedure and haemoptysis after the intervention, with all cases graded as mild to moderate. Although haemoptysis after the intervention showed a nonsignificant trend towards higher occurrence in the ABC group, all cases were mild and self-limited. The device-related complication was delayed airway restenosis secondary to cryoablation-induced tumour necrosis, which occurred in two patients in the ABC group, and this finding may indicate an enhanced cryoablation effect in the ABC group.

MCAO caused by primary lung cancer or metastatic extrapulmonary malignancies is not uncommon and typically exhibits a higher mortality rate.^{16 17} Therapeutic bronchoscopy using multiple modalities has been confirmed to be effective and safe in alleviating symptoms caused by airway obstruction and improving quality of life.^{18 19} The most commonly used techniques include laser therapy, electrocautery, argon plasma coagulation, cryoablation, mechanical debulking, balloon dilation and airway stenting, with the selection based on the clinical scenario.²⁰ According to previous research,^{14 15 21} the reported airway patency rates after 6 weeks of intervention typically range between 92% and 96%. However, the airway patency rates in both ABC and EC groups were lower than anticipated. In this study, we excluded patients with extrinsic compression stenosis requiring airway stent placement, which is associated with high rates of airway restoration.¹ Moreover, unlike observational or retrospective studies, the strict exclusion of recent anti-tumour therapy and the inclusion of over 50% of stage IV cases may also contribute to the relatively lower airway patency rate in our study. As in previous studies,^{1 18 22} symptoms were significantly improved after intervention. However, there was no difference between the two groups at 3 and 6 weeks after the intervention, which may be related to the short follow-up time.

The incidence of bleeding related to cryoablation has been reported to range from 0.7% to 10.0% overall and

life-threatening haemorrhage is rare.^{2 3 23} In our study, most bleeding events during cryoablation were mild and stopped spontaneously. Only one patient in the EC group experienced moderate bleeding that required local medication. The incidence of postintervention haemoptysis in the ABC group was slightly higher than that in the EC group, while no significant difference was found. All haemoptysis events were mild and self-limited in 48 hours. Therefore, the incidence and severity of bleeding events during and after the intervention was comparable between the two groups, indicating the safety of the ABC system. Notably, two cases of delayed airway restenosis were observed in the ABC group. Cryoablation is characterised by multiple forms of cell injury lasting for weeks, so repeated bronchoscopy may be required several days later to remove necrotic tissue.²⁴ Compared with the EC group, the ABC system optimised the cryogen, targeted temperature and working area, which enhanced tumour cell damage and death, and tissue necrosis. Importantly, while these changes may increase the risk of airway restenosis, potentially superior disease control efficacy might be indicated.

Cryoablation has been recommended for the palliative treatment of MCAO, treatment of low-grade airway malignant lesions and early cancer.²⁴⁻²⁶ At present, there are two kinds of devices available for cryoablation: cryoprobe and cryo-spray. The rapid gas conversion creates a dramatic temperature drop to below -40°C at the cryoprobe tip within 1–2 s (-35°C to -50°C for CO_2). The area of tissue injury is estimated visually using the size of the ice formation within the tissue being treated, and the depth of injury is reported to be less than 2 mm with a 120 s freezing time for airway wall via CO_2 cryoprobe.²⁷ Usually, lowering the temperature to -40°C is needed to ensure the adequate tumour cell death. The rate of cooling is most rapidly adjacent to the surface of the cryoablation instrument and lower in the periphery of the ice ball.^{5 28} That's to say, when using the traditional cryoprobe, the ablation effect of the periphery tissue of the freezing area may not achieve as expected, necessitating multiple overlapping freezing sites to compensate, which increases the operation time. Cryo-spray, approved in 2012 by the U.S. Food and Drug Administrations (FDA), is a non-contact modality delivering liquid nitrogen through a flexible catheter to the targeted tissue, achieving rapid cooling to -196°C . This method allows for a larger area (2–3 cm) and lower temperature, optimising the efficacy of ablation.^{20 29 30} It is worth noting that the conversion of liquid nitrogen to gas will result in great volume expansion, so sufficient ventilation is required during the procedure to avoid barotrauma and severe airway obstruction ($>90\%$) or stenosis distal to the bronchus intermedius or left main-stem bronchus is not recommended for cryo-spray.¹¹

The ABC system is an internationally pioneering technology using a balloon as the cryoprobe, which is available in multiple sizes designed for all levels of central airway. The balloon enables simultaneous coverage of extensive lesion areas, and liquid nitrogen circulates internally within a closed-loop system with no leakage or requirement for passive ventilation. This new system offers a lower temperature and expanded freezing area, and it is also safe for use in the distal lobar bronchus. In this study, the duration of the cryoablation procedure was confirmed to be reduced by nearly half in the ABC group. In addition, approximately 40 patients with obstruction distal to bronchus intermedius or left main-stem bronchus were enrolled in the ABC group, indicating the safe use of the ABC system in these anatomical sites.

To our knowledge, this study is one of the few prospective, randomised, controlled studies to explore cryoablation

approach for MCAO, providing a promising alternative in bronchoscopic cryoablation. However, our study also has some limitations. The follow-up period was 6 weeks after intervention, so it is unclear whether the therapeutic advantage could be translated to long-term benefit in symptom control, quality of life or survival. Factors such as performance status, anatomic characteristics of airway stenosis, tumour type, stage and antitumour therapy might influence the prognosis.^{31 32} Therefore, further research is needed to analyse confounding factors and identify the most suitable candidates for this system and optimise perioperative management. Additionally, our study allowed multimodalities for airway tumour debulking before cryoablation, with cryoablation serving only to freeze and thaw residual tumour tissue; it cannot be used to illustrate the clinical efficacy of cryoablation alone. In clinical practice, it should be considered as part of a multimodal interventional regimen. Besides, the exact parameters of cryoablation need to be refined according to different lesions.

CONCLUSIONS

As a part of multimodal interventions in MCAO management, the novel ABC system exhibited non-inferior and even superior effect in airway patency rate compared with the traditional CO₂-driven cryoprobe. Additionally, the ABC system achieved comparable safety and shorter duration of cryoablation procedure. These findings suggest that the ABC system may serve as a promising alternative to traditional flexible cryoprobe in bronchoscopic cryoablation. Further research is needed to elucidate potential confounding factors, optimise individual parameter settings and assess long-term effects, thereby maximising the clinical utility of the ABC system in patients with MCAO.

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and MY performed the therapeutic bronchoscopic procedures and collected data. YS, XZ and SL revised the manuscript. All authors approved the final manuscript for publication. SL supervised the study and is the guarantor of the content of the manuscript.

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